Product Requirements Document

Treatment Plan Timeliness Tracker for Clinical Notes Analyzer

1. Product Name

Working Product Name: IZ Clinical Notes Analyzer – Treatment Plan Timeliness Tracker

2. Product Purpose

The purpose of this product is to help R3 Recovery Services track whether treatment plans for active clients are completed within required timelines, with particular focus on:

1. Initial Treatment Plans completed on admission day.
2. Master Treatment Plans completed within 30 days of admission.
3. Ongoing Treatment Plan Reviews completed on the correct recurring schedule based on level of care.
4. Immediate Treatment Plan Reviews when a client changes level of care.
5. Clear dashboard visibility into compliant, due-soon, urgent, and overdue treatment plan work.

The immediate product goal is not to analyze every aspect of clinical note quality. The first useful version should focus on **date-based treatment plan timeliness** because that is the area the office manager currently tracks manually and repeatedly.

3. Background and Business Context

The office manager currently reviews client charts in Alleva and manually tracks treatment plan due dates in Asana. The process requires checking multiple chart locations:

1. The **Treatment Plan** tab for initial and master treatment plans.
2. The bottom/signature section of treatment plans to confirm signature dates.
3. The **Treatment Plan Review** tab for ongoing treatment plan reviews.
4. The **Level of Care** tab to determine whether the client is in PHP, IOP-5, IOP-19, IOP-3, OP, or has changed levels of care.
5. The treatment plan review signature date to determine the next due date.
6. A separate Asana task or date field to manually track the next due date.

This is time-consuming, error-prone, and difficult to scale across approximately 60 active clients. The office manager needs a system that can be trusted to calculate due dates and surface the right clients at the right time.

4. Product Objectives

4.1 Primary Objectives

The product must:

1. Track all active clients.
2. Determine each client’s current treatment plan timeline status.
3. Calculate the next treatment plan due date.
4. Alert staff before a plan becomes late.
5. Show a simple dashboard of due-soon and overdue treatment plans.
6. Provide an overall compliance percentage for the program.
7. Reduce or eliminate manual Asana-based tracking for treatment plan dates.
8. Be installable and usable by non-technical Windows 10 Home and Windows 11 Home users.

4.2 Secondary Objectives

The product should:

1. Preserve enough audit evidence to explain why a client is compliant or overdue.
2. Make the calculation logic transparent to non-technical users.
3. Allow office staff to manually correct or confirm extracted data when the system cannot determine a date with confidence.
4. Support future expansion into deeper treatment plan content checks.

5. Users and Personas

5.1 Office Manager

Primary user: Marleigh / Office Manager

Needs:

1. See all active clients at a glance.
2. Know which treatment plans are due soon or overdue.
3. Trust the due-date calculation.
4. Avoid duplicate manual tracking in Asana.
5. Review exceptions quickly.
6. Identify therapists/counselors who need follow-up.
7. Understand why the system marked something compliant, due soon, urgent, or overdue.

5.2 Counselor / Therapist

Needs:

1. Know which client treatment plans need to be completed.

2. Understand whether the due date is based on PHP 30-day rules, IOP/OP 60-day rules, admission date, or level-of-care change.
3. See whether a treatment plan has missing signatures.
4. Respond to returned or flagged items.

5.3 Administrator

Needs:

1. Install and configure the app.
2. Manage users and roles.
3. Configure rules if needed.
4. Run tests/checks.
5. Back up local data.
6. Uninstall, repair, or modify the installation without developer assistance.

6. Scope

6.1 In Scope for First Release

The first release must focus on Treatment Plan Timeliness Tracking.

It must support:

1. Active client list tracking.
2. Admission date capture.
3. Current level-of-care capture.
4. Level-of-care history capture, if available.
5. Initial Treatment Plan due-date validation.
6. Master Treatment Plan due-date validation.
7. Ongoing Treatment Plan Review due-date validation.
8. PHP 30-day update cycle.
9. IOP-5, IOP-19, IOP-3, and OP 60-day update cycle.
10. Immediate update requirement after level-of-care change.
11. Staff/client signature date checks where required.
12. Therapist signature date checks for treatment plan reviews.
13. Dashboard with compliance status.
14. Alerts at 10 days before due date, 5 days before due date, and due date.
15. Non-technical Windows install/uninstall/modify support.

6.2 Out of Scope for First Release

The first release should not attempt to fully analyze:

1. Treatment plan clinical content quality.
2. Medical necessity narratives.
3. Diagnosis accuracy.

4. Full payer compliance requirements beyond date/timeliness rules.
5. Automated writeback to Alleva.
6. Full replacement of Alleva.
7. Full replacement of Asana for every workflow outside treatment plan tracking.
8. AI-based clinical judgment.

6.3 Future Scope

Future releases may add:

1. Treatment plan content completeness.
2. Goal/objective/intervention quality checks.
3. Payer-specific rule profiles.
4. Counselor-level performance reporting.
5. Automated reminders by email.
6. Import from Alleva API if official API access is available.
7. Export to CSV/Excel.
8. Optional integration with Asana or other task tools.
9. Broader chart audit workflows beyond treatment plans.

7. Source Workflow Observed from Office Manager

7.1 Initial Treatment Plan

The initial treatment plan is located in the **Treatment Plan** tab.

The system must check whether:

1. An initial treatment plan exists.
2. It was completed on the admission date.
3. It was signed by staff.
4. It was signed by the client.
5. The signature date matches the admission date.

Example observed workflow:

1. Client admission date: February 26.
2. Treatment plan signed on February 26.
3. Because the treatment plan was signed on the admission date, it counts as the initial treatment plan.

7.2 Master Treatment Plan

The master treatment plan is also located in the **Treatment Plan** tab.

The system must check whether:

1. A master treatment plan exists.
2. It was completed within 30 days of admission.

3. It was signed by the therapist/staff.
4. It was signed by the client.
5. The signature date is within 30 days of admission.

Example observed workflow:

1. Admission date: February 26.
2. Master treatment plan created and signed March 3.
3. March 3 is within 30 days of February 26.
4. The client and therapist both signed the plan.
5. The master treatment plan is compliant.

7.3 Ongoing Treatment Plan Reviews

Ongoing reviews are tracked in the **Treatment Plan Review** tab.

The system must check:

1. Whether a treatment plan review exists.
2. The therapist/staff signature date.
3. The due date shown in the chart, when available.
4. The client's level of care at the time of the review.
5. Whether the next due date should be 30 or 60 days from the therapist signature date.

Important office-manager guidance:

1. The client does not always have to sign treatment plan reviews.
2. For current first-release tracking, the key date is the **therapist signature date**.
3. The system should use the therapist signature date as the basis for calculating the next due date.
4. The system should compare the chart's displayed next due date against the calculated due date when available.

7.4 Level of Care

The system must use the **Level of Care** tab to understand:

1. The client's current level of care.
2. Historical level-of-care date ranges.
3. Whether a level-of-care change occurred.
4. Whether a level-of-care row has no discharge/end date, which implies the client is currently in that level of care.

Example observed workflow:

1. Client was in PHP.
2. Client stepped down to IOP-5 on March 30.
3. Treatment plan review was completed on April 2.
4. The review likely reflected the level-of-care change.

5. Since the client is now in IOP-5, the next recurring due date should be based on the 60-day cycle.

8. Business Rules

8.1 Client Scope Rules

Rule ID	Rule	Requirement
BR-001	Active clients only	The tracker must focus on active clients.
BR-002	Discharged clients	Discharged/inactive clients should not appear in active due-soon or overdue queues unless filtered explicitly.
BR-003	Current level of care	A client's current level of care is the active level-of-care row with no discharge/end date, or the most recent level-of-care row if the data does not explicitly mark current status.

8.2 Initial Treatment Plan Rules

Rule ID	Rule	Requirement
BR-010	Initial Treatment Plan required	Every client must have an Initial Treatment Plan.
BR-011	Initial Treatment Plan due date	Initial Treatment Plan is due on Day 1, which is the admission date.
BR-012	Initial staff signature	Initial Treatment Plan must be signed by staff/therapist on the admission date.
BR-013	Initial client signature	Initial Treatment Plan must be signed by the client on the admission date.
BR-014	Initial status	If missing, unsigned, or signed after admission date, status is noncompliant unless manually overridden.

8.3 Master Treatment Plan Rules

Rule ID	Rule	Requirement
BR-020	Master Treatment Plan required	Every client must have a Master Treatment Plan.
BR-021	Master due date	Master Treatment Plan is due within 30 calendar days of admission.

Rule ID	Rule	Requirement
BR-022	Master staff signature	Master Treatment Plan must be signed by therapist/staff within 30 days of admission.
BR-023	Master client signature	Master Treatment Plan must be signed by the client within 30 days of admission.
BR-024	Master status	If missing, unsigned, or signed after the 30-day window, status is noncompliant unless manually overridden.

8.4 Ongoing Review Rules

Rule ID	Rule	Requirement
BR-030	Ongoing review required	After the initial and master treatment plans, ongoing Treatment Plan Reviews must continue while the client is active.
BR-031	Review source tab	Ongoing review data should be pulled from the Treatment Plan Review tab.
BR-032	Key review date	The therapist/staff signature date is the authoritative review completion date for first-release calculations.
BR-033	Client signature optional	Client signature is not required for ongoing review timeliness in the first release.
BR-034	Next due date basis	Next due date is calculated from the last valid treatment plan review therapist signature date.

8.5 Level-of-Care Recurrence Rules

Rule ID	Level of Care	Due Cycle
BR-040	PHP	Every 30 days
BR-041	IOP-5	Every 60 days
BR-042	IOP-19	Every 60 days
BR-043	IOP-3	Every 60 days
BR-044	OP	Every 60 days

The system must normalize level-of-care labels. For example, `IOP5`, `IOP-5`, `IOP 5`, and `IOPF` may need to be mapped to the same configured level if R3 confirms they are equivalent.

8.6 Level-of-Care Change Rules

Rule ID	Rule	Requirement
BR-050	Immediate update on LOC change	A treatment plan update is due immediately when the client changes level of care.
BR-051	LOC change detection	The system must detect new level-of-care rows or changed level-of-care effective dates.
BR-052	LOC review evidence	If a treatment plan review occurs shortly after a level-of-care change, the system should link the review to that change.
BR-053	LOC current-cycle reset	After a level-of-care change review is completed, the next recurring due date should be based on the new level of care.
BR-054	PHP to IOP example	If a client moves from PHP to IOP-5, the review after the change should trigger a 60-day next due date instead of a 30-day PHP cycle.

8.7 Alert Rules

Rule ID	Alert	Timing	Display
BR-060	Compliant	More than 10 days before due date	Green
BR-061	Due Soon Warning	10 days before due date	Yellow
BR-062	Urgent Warning	5 days before due date	Yellow/orange
BR-063	Due Date / Overdue	On due date and after	Red
BR-064	Missing data	Unable to calculate due date	Gray or purple exception state

Note: Marleigh used “Day of due date – overdue.” The product should support that behavior exactly unless R3 later decides to distinguish “Due Today” from “Overdue.”

8.8 Next Due Date Calculation Rules

Rule ID	Scenario	Calculation
BR-070	PHP ongoing review	Last therapist signature date + 30 calendar days
BR-071	IOP-5 ongoing review	Last therapist signature date + 60 calendar days
BR-072	IOP-19 ongoing review	Last therapist signature date + 60 calendar days
BR-073	IOP-3 ongoing review	Last therapist signature date + 60 calendar days
BR-074	OP ongoing review	Last therapist signature date + 60 calendar days
BR-075	Level-of-care change	Due immediately on or after effective level-of-care change date

Rule ID	Scenario	Calculation
BR-076	Master Treatment Plan	Admission date + 30 calendar days
BR-077	Initial Treatment Plan	Admission date

9. Data Requirements

9.1 Client-Level Data

The system must store or ingest:

1. Client ID / patient ID.
2. Client name, if permitted.
3. Active/inactive/discharged status.
4. Admission date.
5. Discharge date, if applicable.
6. Current level of care.
7. Counselor/therapist.
8. Program/location, if needed.
9. Last updated/imported timestamp.

9.2 Treatment Plan Data

The system must store or ingest:

1. Treatment plan type:

2. Initial Treatment Plan.

3. Master Treatment Plan.
4. Treatment Plan Review.
5. Created date.
6. Staff/therapist signature date.
7. Staff/therapist signer name.
8. Client signature date, where applicable.
9. Client signer name, where applicable.
10. Displayed next due date, if present in Alleva.
11. Review content metadata, if available.
12. Source tab/location.
13. Source document ID, if available.
14. Import/extraction confidence.

9.3 Level-of-Care Data

The system must store or ingest:

1. Level of care name/code.
2. Level-of-care start/effective date.
3. Level-of-care discharge/end date.
4. Current level-of-care indicator.
5. Prior level of care.
6. Change date.
7. Source tab/location.
8. Import/extraction confidence.

9.4 Derived Fields

The system must calculate:

1. Initial Treatment Plan due date.
2. Initial Treatment Plan actual completion date.
3. Initial Treatment Plan status.
4. Master Treatment Plan due date.
5. Master Treatment Plan actual completion date.
6. Master Treatment Plan status.
7. Last valid treatment plan review date.
8. Applicable recurrence interval.
9. Next treatment plan due date.
10. Days until due.
11. Alert status.
12. Overall client treatment plan status.
13. Program compliance percentage.
14. Evidence summary.
15. Exception reason, if unable to evaluate.

10. Status Model

Each active client should have one current treatment plan tracking status.

10.1 Status Values

Status	Meaning	Color
Compliant	Treatment plan is current and more than 10 days from due date	Green
Due Soon	Treatment plan is due within 10 days	Yellow
Urgent	Treatment plan is due within 5 days	Yellow/orange
Overdue	Treatment plan is due today or past due	Red

Status	Meaning	Color
Missing Data	System cannot calculate because key data is missing	Gray/purple
Needs Review	System found conflicting evidence	Gray/purple

10.2 Status Priority

If multiple conditions apply, the system should use the most severe status:

1. Overdue.
2. Urgent.
3. Due Soon.
4. Missing Data / Needs Review.
5. Compliant.

11. Dashboard Requirements

11.1 Dashboard Summary Cards

The dashboard must show:

1. Total active clients.
2. Number compliant.
3. Number due soon.
4. Number urgent.
5. Number overdue.
6. Number missing data / needs review.
7. Overall program compliance percentage.

11.2 Compliance Percentage

The first-release compliance percentage should be:

`Compliant active clients ÷ active clients with enough data to evaluate`

The dashboard should also separately show clients excluded from the denominator because of missing data.

Alternative formula for later review:

`Compliant active clients ÷ all active clients`

The product should make clear which formula is being used.

11.3 Main Dashboard Table

The dashboard table must include:

1. Client name or patient ID.
2. Current level of care.
3. Counselor/therapist.
4. Admission date.
5. Last treatment plan date.
6. Next due date.
7. Days until due.
8. Status.
9. Evidence/source.
10. Last imported/checked timestamp.
11. Action button: View Details.

11.4 Dashboard Filters

The dashboard should support filters for:

1. All active clients.
2. Due soon.
3. Urgent.
4. Overdue.
5. Missing data / needs review.
6. Level of care.
7. Counselor/therapist.
8. Program.
9. Admission date range.

11.5 Dashboard Sorting

The table should default-sort by:

1. Overdue first.
2. Urgent second.
3. Due soon third.
4. Nearest due date first within each status group.

12. Client Detail Page Requirements

For each client, the detail page must show:

1. Client overview:

2. Patient ID.

3. Name, if allowed.
4. Active status.
5. Admission date.
6. Current level of care.
7. Counselor/therapist.
8. Treatment plan timeline:
9. Admission date.
10. Initial Treatment Plan due date.
11. Initial Treatment Plan actual signature date.
12. Master Treatment Plan due date.
13. Master Treatment Plan actual signature date.
14. Level-of-care changes.
15. Treatment Plan Review dates.
16. Calculated next due date.
17. Current compliance result:
18. Status.
19. Due date.
20. Days until due.
21. Rule used.
22. Evidence.
23. Confidence.
24. Any missing data.
25. Source evidence:
26. Source tab.
27. Source document.
28. Signature date.
29. Staff/client signature evidence.
30. Level-of-care evidence.
31. Manual override section:
32. Override status.
33. Override due date.
34. Override reason.

35. User making override.
36. Timestamp.
37. Audit log entry.

13. Alert Requirements

13.1 In-App Alerts

The app must show due alerts in the dashboard.

Alert levels:

1. 10 days before due date: warning.
2. 5 days before due date: urgent.
3. On due date: overdue/red.
4. After due date: overdue/red.

13.2 Future Email Alerts

The first release may only require in-app alerts. A future release should support email alerts.

Potential email alert recipients:

1. Office manager.
2. Assigned counselor/therapist.
3. Admin users.

Email alert frequency should be configurable.

13.3 Alert Suppression

The system should avoid duplicate/noisy alerts by tracking:

1. Alert generated date.
2. Alert level.
3. Client.
4. Due date.
5. Whether the alert has already been acknowledged.

14. Data Ingestion Requirements

14.1 First Release Ingestion

Because the current implementation is upload-first, the first release should support one or more of:

1. Manual upload of exported documents.
2. Upload of CSV/JSON export files.

3. Manual entry/editing of key fields.
4. Structured import template for active clients and treatment plan history.

14.2 Future Alleva Integration

The product should preserve a future path to Alleva API/FHIR integration, but live import should remain gated until:

1. R3 has official API access.
2. Alleva confirms endpoints and authentication method.
3. Tenant base URL is known.
4. Attachment behavior is documented.
5. Pagination/rate limits are documented.
6. Sandbox patients are available.
7. Compliance/legal approval exists.
8. BAA/security terms are complete.

14.3 Manual Data Correction

The app must allow authorized users to correct:

1. Admission date.
2. Level of care.
3. Treatment plan signature date.
4. Next due date.
5. Active/discharged status.
6. Counselor assignment.
7. Missing or misclassified treatment plan type.

Every correction must be audited.

15. Installation, Uninstallation, and Modification Requirements

15.1 Target Operating Systems

The app must be compatible with:

1. Windows 10 Home.
2. Windows 11 Home.

It should not require:

1. Docker.
2. PostgreSQL installation.
3. Developer tools.
4. Git.
5. Node.js.
6. Command-line usage by ordinary users.

7. Admin-level technical knowledge.

15.2 Installer Requirements

The product should provide a non-technical installer, such as a signed `.exe` or `.msi`.

The installer must:

1. Install the app in a standard Windows application location.
2. Create a Start Menu shortcut.
3. Optionally create a Desktop shortcut.
4. Install or bundle the required runtime.
5. Include the built frontend assets.
6. Configure local app-data storage.
7. Generate local secrets on first run.
8. Set up local SQLite database.
9. Create or display bootstrap admin credentials securely.
10. Validate the installation after completion.
11. Offer to launch the app after installation.

15.3 First-Run Experience

On first run, the app should:

1. Check runtime readiness.
2. Confirm local database availability.
3. Confirm upload directory availability.
4. Confirm encryption key exists.
5. Confirm rules file is valid.
6. Show the local app URL.
7. Prompt for initial admin password change if appropriate.
8. Warn if the app is being run from a cloud-synced folder.
9. Show a simple “Ready / Needs Attention” screen.

15.4 Modify Installation

The Windows installer must support “Modify” or “Repair” behavior.

Modify/repair should allow:

1. Repair app binaries.
2. Recreate shortcuts.
3. Reinstall bundled runtime.
4. Rebuild or replace frontend assets.
5. Preserve local database.
6. Preserve encrypted uploads.
7. Preserve `.env` encryption keys.
8. Preserve user accounts.

9. Preserve audit logs unless the user explicitly chooses otherwise.

15.5 Uninstall Requirements

The uninstaller must:

1. Remove application binaries.
2. Remove Start Menu/Desktop shortcuts.
3. Stop any local running process.
4. Ask whether to preserve or delete local app data.
5. Clearly warn that deleting local app data may delete:
6. SQLite database.
7. Encrypted uploads.
8. Audit logs.
9. `.env` encryption key.
10. Saved API configuration.
11. Default to preserving local app data.
12. Provide a “delete all local data” option only after explicit confirmation.
13. Produce an uninstall log.

15.6 Upgrade Requirements

The installer/updater must:

1. Preserve local data.
2. Preserve encryption keys.
3. Run schema compatibility checks.
4. Back up the local database before schema changes.
5. Validate rules after upgrade.
6. Show success/failure in plain English.
7. Provide rollback instructions if upgrade fails.

16. Security and Privacy Requirements

16.1 Local Data Protection

The app must:

1. Store clinical documents encrypted at rest.
2. Store API keys and secrets encrypted.
3. Keep runtime data outside the source code folder.
4. Avoid storing PHI in logs.
5. Avoid storing PHI in YAML rule files.
6. Require login before viewing client information.
7. Enforce role-based access.

16.2 Audit Requirements

The app must audit:

1. Login success/failure.
2. User lockouts.
3. User creation/update/deactivation.
4. Uploads.
5. Downloads.
6. Treatment plan calculations.
7. Manual overrides.
8. Rule changes.
9. Settings changes.
10. API configuration changes.
11. Approvals/returns by office manager.

16.3 Backup Requirements

The product must document and preferably automate backup of:

1. SQLite database.
2. Encrypted uploads.
3. `.env` file containing encryption key.
4. Audit logs.
5. Rules configuration.

The backup workflow must warn that losing the encryption key can make stored clinical documents unrecoverable.

17. Functional Requirements

17.1 Client Tracking

ID	Requirement	Priority
FR-001	System shall maintain a list of active clients.	Must
FR-002	System shall allow filtering by active/discharged status.	Must
FR-003	System shall show current level of care for each active client.	Must
FR-004	System shall show assigned counselor/therapist when available.	Should

17.2 Treatment Plan Timeline

ID	Requirement	Priority
FR-010	System shall identify Initial Treatment Plan.	Must

ID	Requirement	Priority
FR-011	System shall validate Initial Treatment Plan completed on admission date.	Must
FR-012	System shall identify Master Treatment Plan.	Must
FR-013	System shall validate Master Treatment Plan completed within 30 days of admission.	Must
FR-014	System shall identify ongoing Treatment Plan Reviews.	Must
FR-015	System shall use therapist signature date as ongoing review completion date.	Must
FR-016	System shall calculate next due date from last review date and current level of care.	Must
FR-017	System shall detect missing or conflicting treatment plan dates.	Must

17.3 Level of Care

ID	Requirement	Priority
FR-020	System shall track level-of-care history.	Must
FR-021	System shall determine current level of care.	Must
FR-022	System shall apply 30-day recurrence for PHP.	Must
FR-023	System shall apply 60-day recurrence for IOP-5, IOP-19, IOP-3, and OP.	Must
FR-024	System shall detect level-of-care changes.	Should
FR-025	System shall flag treatment plan review due immediately after level-of-care change.	Must

17.4 Alerts and Dashboard

ID	Requirement	Priority
FR-030	System shall show Due Soon clients 10 days before due date.	Must
FR-031	System shall show Urgent clients 5 days before due date.	Must
FR-032	System shall show clients as Overdue on the due date and after.	Must
FR-033	System shall color-code compliant/due-soon/overdue clients.	Must
FR-034	System shall display overall program compliance percentage.	Must
FR-035	System shall allow sorting by due date and status.	Must
FR-036	System shall allow filtering by status, level of care, and counselor.	Should

17.5 Manual Review and Override

ID	Requirement	Priority
FR-040	System shall allow authorized users to manually correct extracted dates.	Must
FR-041	System shall require a reason for manual override.	Must
FR-042	System shall audit all overrides.	Must
FR-043	System shall show original extracted value and override value.	Should

17.6 Installation and Maintenance

ID	Requirement	Priority
FR-050	System shall provide a non-technical Windows installer.	Must
FR-051	Installer shall support Windows 10 Home and Windows 11 Home.	Must
FR-052	Installer shall not require Docker, Git, Node.js, or PostgreSQL.	Must
FR-053	Installer shall create Start Menu and optional Desktop shortcuts.	Must
FR-054	Installer shall support repair/modify installation.	Must
FR-055	Uninstaller shall preserve user data by default.	Must
FR-056	Uninstaller shall require explicit confirmation before deleting local app data.	Must

18. Non-Functional Requirements

18.1 Usability

The app must be usable by a non-technical office manager.

Design principles:

1. Plain-English status messages.
2. Minimal technical terminology.
3. Clear colors and labels.
4. One-click dashboard filters.
5. Obvious “why is this overdue?” explanation.
6. Simple install/uninstall/repair experience.

18.2 Reliability

The app must:

1. Start consistently on Windows 10 Home and Windows 11 Home.
2. Validate readiness at startup.

3. Fail safely if rules are invalid.
4. Avoid corrupting local data on crash.
5. Back up database before upgrades.
6. Preserve encrypted uploads.

18.3 Performance

The app should support at least:

1. 100 active clients.
2. 1,000 historical treatment plan records.
3. 40 uploaded files per binder.
4. Dashboard load under 3 seconds for typical local use.
5. Client detail load under 3 seconds for typical local use.

18.4 Maintainability

The app should keep treatment plan rules in a configurable rules file where possible.

The rule logic should be:

1. Transparent.
2. Versioned.
3. Testable.
4. Not dependent on an LLM.
5. Easy to update when R3 requirements change.

19. User Experience Requirements

19.1 Main Navigation

Primary navigation should include:

1. Dashboard.
2. Active Clients.
3. Due Soon.
4. Overdue.
5. Upload/Import.
6. Review Queue.
7. Reports.
8. Settings.
9. Users/Admin.
10. Audit Logs.

19.2 Dashboard Visual Design

Color coding:

1. Green = compliant.
2. Yellow = due soon.
3. Orange/yellow = urgent.
4. Red = overdue.
5. Gray/purple = missing data or needs review.

Each row should include a plain-English explanation, such as:

1. “Next review due in 8 days based on IOP-5 60-day rule.”
2. “Overdue: PHP review was due 05/29/2026 based on 04/29/2026 therapist signature.”
3. “Needs review: current level of care could not be determined.”

19.3 Client Detail Timeline

The client detail page should display a timeline:

1. Admission date.
2. Initial treatment plan.
3. Master treatment plan.
4. Level-of-care changes.
5. Treatment plan reviews.
6. Next due date.
7. Alert status.

20. Reporting Requirements

First-release reports:

1. Current active-client compliance report.
2. Due-soon report.
3. Overdue report.
4. Missing-data report.
5. Counselor-level summary.
6. Export to CSV.

Future reports:

1. Monthly compliance trend.
2. Level-of-care compliance comparison.
3. Counselor caseload risk.
4. Audit history report.
5. Treatment plan completion aging.

21. Acceptance Criteria

21.1 Initial Treatment Plan

Given a client admitted on February 26,
when the system finds an initial treatment plan signed by both staff and client on February 26,
then the Initial Treatment Plan status is compliant.

Given a client admitted on February 26,
when the system finds no initial treatment plan or the plan is signed after February 26,
then the Initial Treatment Plan status is noncompliant.

21.2 Master Treatment Plan

Given a client admitted on February 26,
when the system finds a master treatment plan signed by both therapist and client on March 3,
then the Master Treatment Plan status is compliant.

Given a client admitted on February 26,
when the master treatment plan is signed more than 30 days after admission,
then the Master Treatment Plan status is noncompliant.

21.3 PHP Review Cycle

Given a client currently in PHP,
and the last treatment plan review therapist signature date is April 2,
then the next treatment plan due date is May 2.

21.4 IOP-5 Review Cycle

Given a client currently in IOP-5,
and the last treatment plan review therapist signature date is April 2,
then the next treatment plan due date is June 1.

21.5 Level-of-Care Change

Given a client moves from PHP to IOP-5 on March 30,
when a treatment plan review is completed on April 2,
then the system should treat April 2 as the review date after the level-of-care change and calculate the next due date using the IOP-5 60-day rule.

21.6 Alerts

Given a treatment plan due date is 10 days away,
then the client appears in Due Soon.

Given a treatment plan due date is 5 days away,
then the client appears in Urgent.

Given a treatment plan due date is today or earlier,
then the client appears in Overdue.

21.7 Dashboard

Given 60 active clients,
when the dashboard loads,
then it shows counts for compliant, due soon, urgent, overdue, and missing data clients, plus overall compliance percentage.

21.8 Installation

Given a non-technical user on Windows 11 Home,
when they run the installer,
then the app installs, creates shortcuts, configures local storage, starts successfully, and opens in a browser without requiring Docker, Git, Node.js, PostgreSQL, or command-line steps.

21.9 Uninstall

Given a user uninstalls the app,
when the uninstall wizard runs,
then it removes the app but preserves local clinical data by default and warns before deleting any local app data or encryption keys.

22. Open Questions

1. What exact Alleva labels distinguish Initial Treatment Plan from Master Treatment Plan?
2. Are `IOPF`, `IOP-5`, and `IOP 5` equivalent in R3 terminology?
3. Does “Day of due date = overdue” mean red on the due date, or should there be a separate “Due Today” status?
4. How many days after a level-of-care change count as “immediate” for the required treatment plan update?
5. Are client signatures always required for Initial and Master Treatment Plans?
6. Are client signatures ever required for ongoing Treatment Plan Reviews in specific programs or payer cases?
7. Should discharged clients be excluded immediately, or kept visible for a grace/audit period?
8. Should the compliance percentage exclude clients with missing data, or count missing data as noncompliant?
9. Should alerts be in-app only for first release, or should email alerts be included immediately?
10. What is the desired source of truth for active client list: manual upload, Alleva export, API, or manual entry?
11. Should the app replace Asana tracking entirely for this workflow, or export dates/tasks into Asana?
12. Who can override due dates: admin only, office manager, or counselor?
13. Should overrides expire or require periodic review?

23. Recommended Release Plan

Phase 1 – Manual/Upload-Based MVP

1. Active client dashboard.
2. Manual or CSV import of client list.
3. Manual or uploaded treatment plan dates.
4. Rules-based due date calculation.
5. Due soon / urgent / overdue dashboard.
6. Compliance percentage.
7. Local Windows installer prototype.
8. Audit log for manual edits and overrides.

Phase 2 – Alleva Export Enhancement

1. More structured import from Alleva exports.
2. Treatment Plan tab extraction.
3. Treatment Plan Review tab extraction.
4. Level-of-care tab extraction.
5. Signature-date extraction.
6. Confidence scoring and missing-data queue.

Phase 3 – Operational Alerts and Reporting

1. Email reminders.
2. Counselor-level reports.
3. CSV/Excel exports.
4. Historical compliance trend.
5. Optional Asana task export/integration.

Phase 4 – API/FHIR Integration

1. Alleva API/FHIR discovery.
2. Read-only patient/document import.
3. DocumentReference/Binary mapping.
4. Automated scheduled refresh.
5. Vendor-approved production integration.

24. Product Success Metrics

The product should be considered successful if:

1. Office manager no longer needs to manually track most treatment plan due dates in Asana.
2. Dashboard correctly identifies due-soon and overdue treatment plans.
3. Staff can explain why every due date was calculated.
4. The system handles all active clients without manual spreadsheet work.
5. The office manager trusts the alerts.

6. The app can be installed and run on Windows 10 Home and Windows 11 Home by a non-technical user.
7. R3 can validate treatment plan timeliness faster than the current manual workflow.